

SQL Review Research Activity

Individual Research and Report Writing

Instructions

Complete an individual report addressing all sections below.

Part 1: SQL Fundamentals Review

Research and explain the following:

1. What is SQL?
2. Why is SQL important in database systems?
3. Explain the difference between:
 - DDL
 - DML
 - DCL
 - TCL
4. Describe the purpose of the following SQL clauses/commands:
 - SELECT
 - WHERE
 - ORDER BY
 - GROUP BY
 - HAVING
 - JOIN

Part 2: Query Analysis

For each query below:

- Explain what the query does
- Describe a real-world use case for the query

Query A

```
SELECT * FROM employees;
```

Query B

```
SELECT department, AVG(salary)
FROM employees
GROUP BY department;
```

Query C

```
SELECT c.name, o.total
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id;
```

Part 3: SQL in Real-World Applications

Choose **three (3) industries** and explain how SQL is used in each.

Possible industries include:

- Banking
- Healthcare
- E-commerce
- Education
- Government
- Logistics
- etc.

For each industry, discuss:

1. What types of data are stored?
2. How SQL is used to manage/analyze the data
3. Example SQL-related tasks in that industry

Part 4: Reflection

Answer the following:

1. Which SQL concept do you find most challenging? Why?
2. Which SQL concept do you believe is most important for industry use? Why?
3. How does SQL support data-driven decision making?

Deliverables

Submit:

- A report in **PDF** and **Word format**

- Minimum length: **2–5 pages**
- Clear headings for each section
- Proper formatting and grammar

Assessment Criteria

Criteria	Marks
Completeness of Research	30
Accuracy of SQL Explanation	30
Depth of Real-World Analysis	20
Reflection Quality	10
Report / Formatting	10

Total: 100 Marks

Submission Deadline: 04th May 2026, 11:59 AM (Today)